

PEM Fuel Cell Stack Modelling & Parameter Tuning

From physical principles to per-cell calibration

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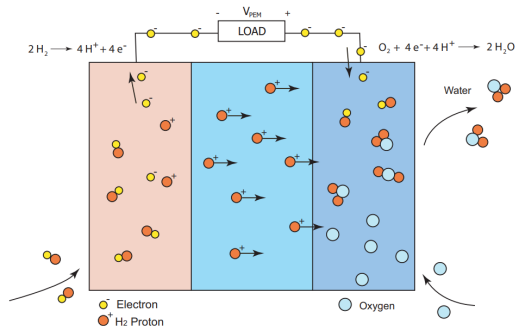
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Outline

- 1 Physical Principle
- 2 Single-Cell Model
- 3 State-space representation
- 4 Experiment
- 5 Optimization
- 6 Results
- 7 Simulation vs Experiment
- 8 Conclusion

Operating Principle of a PEM Fuel Cell

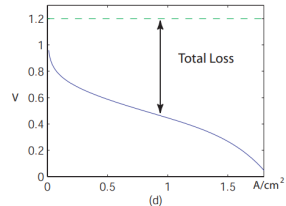
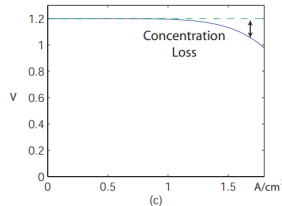
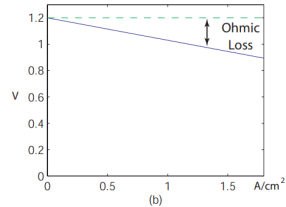
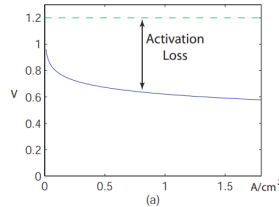
- **Anode:** $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
- **Cathode:** $\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$
- **Overall:**
 $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{electricity} + \text{heat}$
- Proton Exchange Membrane (PEM) conducts H^+ , blocks electrons
- Water management critical: too dry $\rightarrow$ conductivity loss; flooding $\rightarrow$ transport blockage



Voltage Losses — Polarisation Curve

- **Activation loss** — kinetic limitation at low current
- **Ohmic loss** — resistive drop, linear with current
- **Concentration loss** — mass-transport limit at high current

$$V_{fc} = E - V_{act} - V_{ohm} - V_{con}$$



Open-Circuit Voltage (Nernst)

Derived from thermodynamics (Gibbs free energy, Faraday's law):

$$E = 1.229 - 0.85 \times 10^{-3} (T - 298.15) + 4.31 \times 10^{-5} T (\ln P_{H_2} + \frac{1}{2} \ln P_{O_2})$$

- 1.229 V — standard reversible voltage at 25 °C
- -0.85×10^{-3} — temperature coefficient from $\Delta S^0/2F$ (liquid water)
- Last term — Nernst pressure correction

Voltage Loss Equations

Activation loss (kinetics)

$$v_{\text{act}} = \xi_1 + \xi_2 T + \xi_3 T \ln C_{O_2} + \xi_4 T \ln i_{FC}$$

Ohmic loss (membrane resistance)

$$v_{\text{ohm}} = i_{FC} \left(\frac{t_m}{A_{fc} \sigma_m(\lambda_m, T)} + c \right), \quad \sigma_m = (b_{11} \lambda_m - b_{12}) \exp \left[b_2 \left(\frac{1}{303} - \frac{1}{T} \right) \right]$$

Concentration loss (mass transport)

$$v_{\text{con}} = -B \ln \left(1 - \frac{i_{FC}}{i_{\text{max}}} \right)$$

Tuned per cell: $\xi_1, \xi_2, \xi_3, \xi_4, B, c$ **Fixed:** $T, P_{O_2}^* = 0.21 P_{\text{atm}}, \lambda_m, A_{fc}, t_m, i_{\text{max}}$

Membrane Water Content — λ_m

Relative humidities are algebraic outputs of the water mass states:

$$\phi_{an} = \frac{p_{v,an}}{p_{sat}(T)}, \quad p_{v,an} = \frac{m_{w,an} R_v T}{V_{an}} \quad \phi_{ca} = \frac{p_{v,ca}}{p_{sat}(T)}, \quad p_{v,ca} = \frac{m_{w,ca} R_v T}{V_{ca}}$$

Springer sorption isotherm converts activity $a \in [0, 1]$ to water content:

$$\lambda(a) = 0.043 + 17.81 a - 39.85 a^2 + 36.0 a^3$$

Membrane water content is the average of both sides:

$$\lambda_m = \frac{\lambda(\phi_{an}) + \lambda(\phi_{ca})}{2}$$

λ_m then enters the membrane conductivity σ_m and hence v_{ohm} .

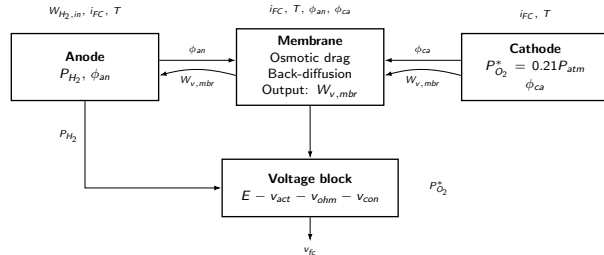
Note: in the simplified model λ_m is held constant (e.g. = 14) when water dynamics are not tracked.

Subsystem Architecture

Three coupled subsystems:

- **Anode** — hydrogen pressure P_{H_2} , anode humidity ϕ_{an}
- **Membrane** — water transport (electro-osmotic drag + back-diffusion)
- **Cathode** — constant $P_{O_2}^*$, cathode humidity ϕ_{ca}

All three feed into the **voltage block** to compute v_{fc}



State-Space Representation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad y = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

$$\mathbf{x} = \begin{bmatrix} P_{H_2} \\ m_{w,an} \\ m_{w,ca} \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} i_{FC} \\ W_{H_2,in} \end{bmatrix}, \quad y = v_{fc}$$

$$\mathbf{A} = \begin{bmatrix} -\frac{k_{an}R_{H_2}T}{V_{an}} & 0 & 0 \\ 0 & -\frac{D_w}{V_{an}t_m} & \frac{D_w}{V_{ca}t_m} \\ 0 & \frac{D_w}{V_{an}t_m} & -\frac{D_w}{V_{ca}t_m} \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} -\frac{M_{H_2}nR_{H_2}T}{2FV_{an}} & \frac{R_{H_2}T}{V_{an}} \\ \frac{n_dM_v}{F} & 0 \\ \frac{(n-2n_d)M_v}{2F} & 0 \end{bmatrix}$$

$\mathbf{C}$, $\mathbf{D}$: Jacobians of v_{fc} w.r.t. $\mathbf{x}$ and $\mathbf{u}$ at the operating point

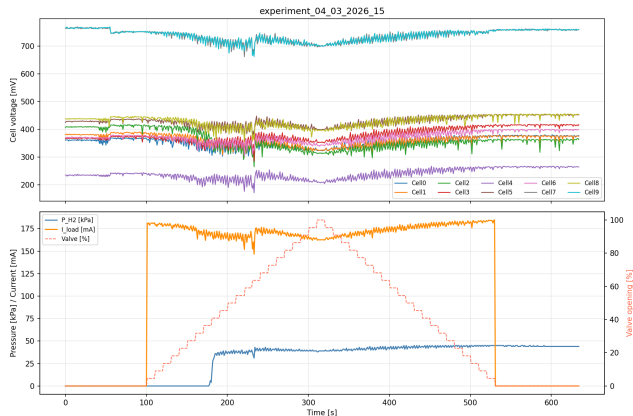
Experimental Setup — Inputs & Outputs

Inputs (controlled):

- H₂ valve opening (0–100 %)
- H₂ inlet pressure (gauge → absolute atm)
- H₂ and air mass flow rates

Outputs (measured):

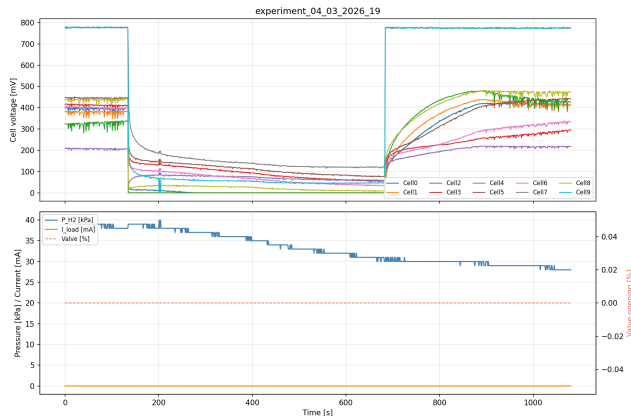
- Individual cell voltages: $v_0 \dots v_9$ [mV]
- Stack voltage V_{stack} [V]
- Load current: $I = V_{\text{stack}}/25 \Omega$ [A]



Experimental Setup — Experiment 19

Key difference vs Exp 15:

- **No valve control** — fixed 25 Ω resistor connected at $t = 135$ s
- Resistor **removed** at $t = 684$ s — current returns to 0
- Current set by Ohm's law:
 $I = V_{\text{stack}}/R$ (loaded window only)
- Longer run: 1081 s vs 635 s
- Cells 0, 1, 2 & 8 drop to ≈ 0 V under load (hardware degradation)



Two-Stage Per-Cell Optimisation

Objective: minimise RMSE between predicted and measured cell voltage

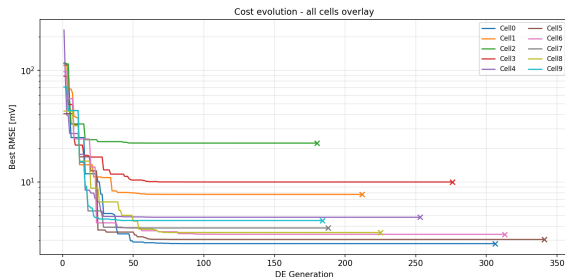
$$\min_{\xi_1, \xi_2, \xi_3, \xi_4, B, c} \sqrt{\frac{1}{N} \sum_{i=1}^N (v_{\text{model}}(I_i, P_{H_2, i}) - v_{\text{meas}, i})^2}$$

- ① **Differential Evolution** (global) — population 20, max 800 generations, seed 42
- ② **L-BFGS-B** (local polish) — refines DE solution within bounds

Physical bounds enforced:

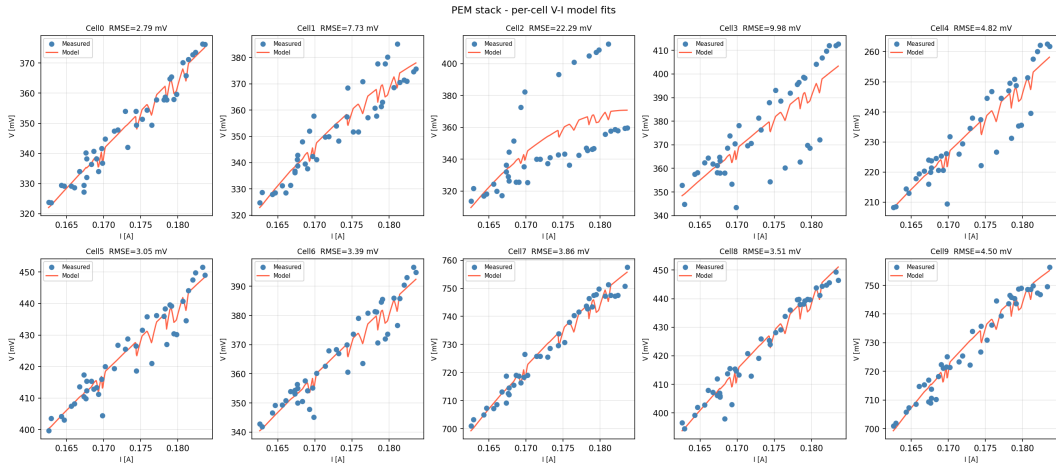
- $\xi_4 \leq 0$ — activation loss increases with current
- $B \geq 0$ — concentration loss is always a drop
- $c \geq 0$ — ohmic resistance is always positive

Cost Evolution



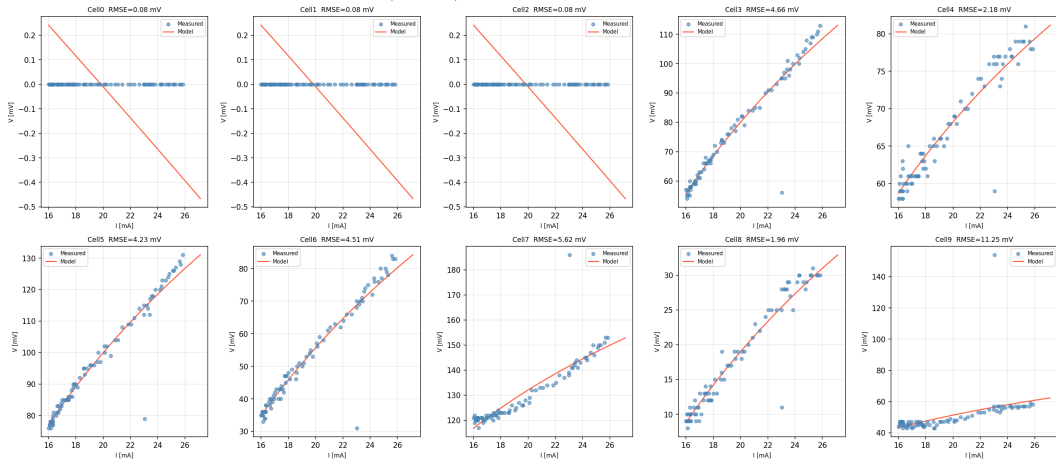
- Most cells converge within **50 generations**
- Cell 2 stagnates at 22 mV — hardest to fit
- L-BFGS-B polish gives negligible further gain
- Cells 1, 3 plateau > 7 mV — model mismatch

V-I Fits: All 10 Cells — Experiment 15



V-I Fits: All 10 Cells — Experiment 19

Experiment 19 — per-cell V-I fits (load detected at $t=135$ s, OCV=4568 mV)

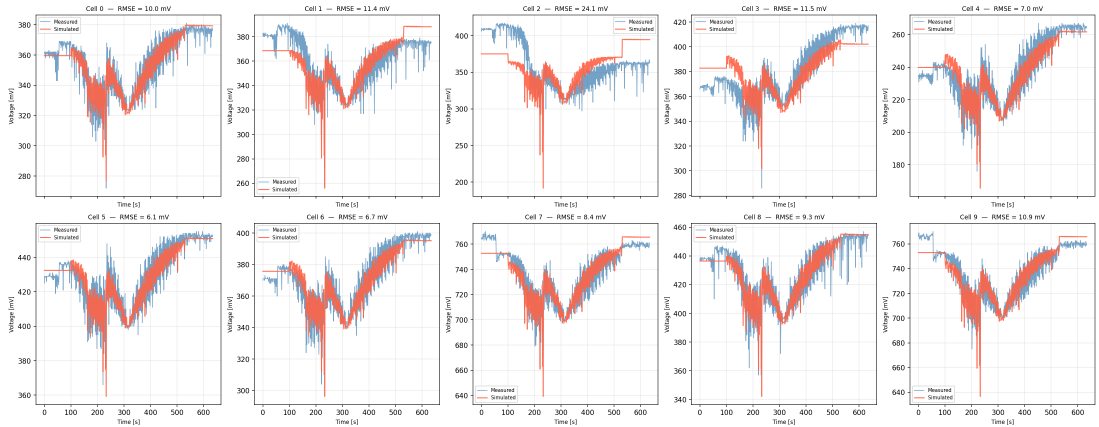


Tuned Parameters & RMSE

Cell	RMSE (mV)	ξ_1	ξ_2	ξ_3	ξ_4	B	c
Cell0	2.8	-0.884	1.06×10^{-2}	5.0×10^{-4}	-1.26×10^{-3}	≈ 0	≈ 0
Cell1	7.7	-7.457	1.11×10^{-3}	-1.0×10^{-3}	-3.32×10^{-3}	0.305	≈ 0
Cell2	22.3	1.350	1.87×10^{-3}	1.3×10^{-3}	-8.07×10^{-3}	1.003	1.1×10^{-6}
Cell3	10.0	-2.885	1.30×10^{-3}	-4.6×10^{-4}	-1.30×10^{-3}	≈ 0	≈ 0
Cell4	4.8	-7.935	1.12×10^{-2}	-8.1×10^{-4}	-1.18×10^{-3}	≈ 0	≈ 0
Cell5	3.0	-7.967	3.28×10^{-3}	-1.3×10^{-3}	-1.14×10^{-3}	≈ 0	≈ 0
Cell6	3.4	1.354	2.53×10^{-2}	1.9×10^{-3}	-1.23×10^{-3}	≈ 0	≈ 0
Cell7	3.9	-6.255	6.39×10^{-3}	-5.2×10^{-4}	-2.53×10^{-3}	0.181	≈ 0
Cell8	3.5	-0.655	2.72×10^{-3}	1.1×10^{-4}	-1.65×10^{-3}	0.043	≈ 0
Cell9	4.5	-3.032	9.24×10^{-3}	3.1×10^{-4}	-2.84×10^{-3}	0.229	≈ 0

Exp 15 — Per-Cell Voltage: Measured vs Simulated

Experiment 04-03-2026 #15 — Measured vs Simulated (tuned semi-empirical model)

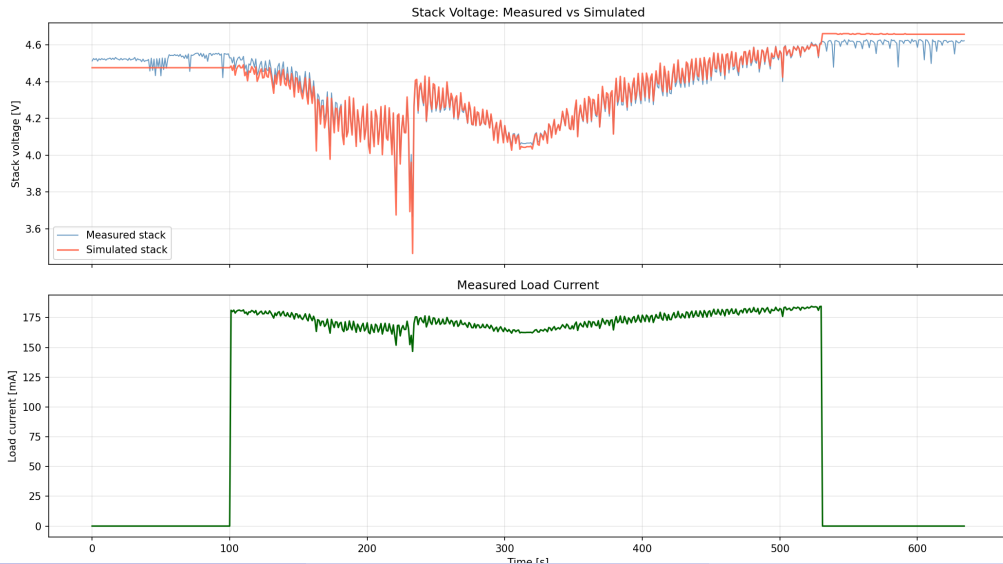


OCV region: Nernst model (T_{eff} fitted per cell).

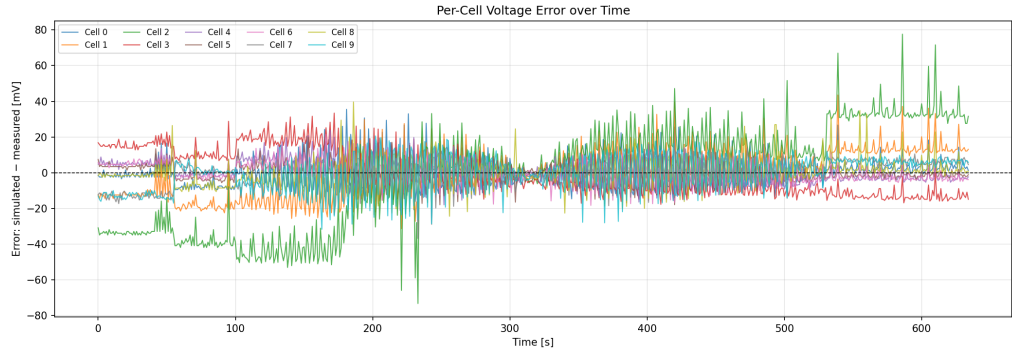
Loaded region: tuned semi-empirical model.

Mean RMSE = 10.6 mV

Exp 15 — Stack Voltage & Load Current



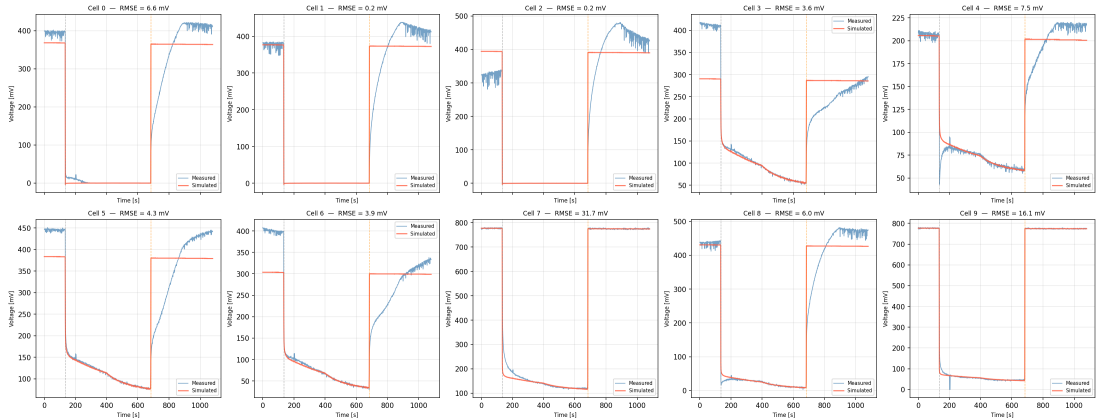
Exp 15 — Per-Cell Voltage Error



Error = simulated – measured. Positive bias in degraded cells (Cell 2).

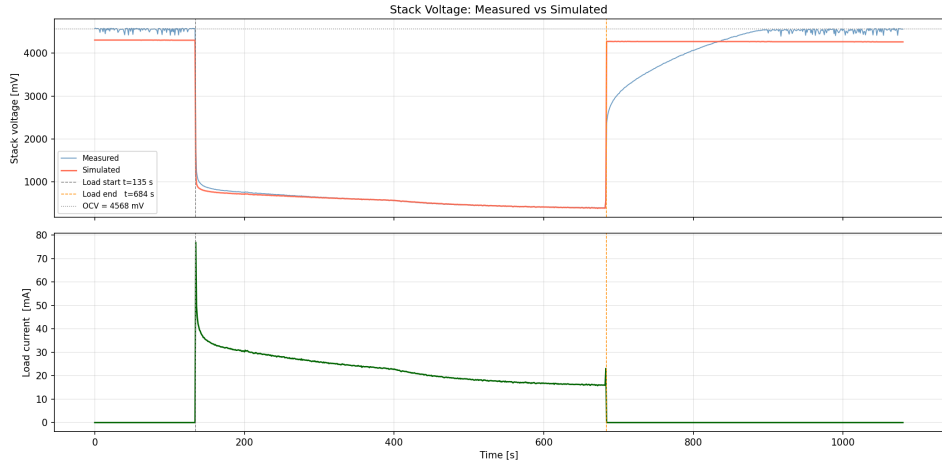
Exp 19 — Per-Cell Voltage: Measured vs Simulated

Experiment 04-03-2026 #19 — Measured vs Simulated
(resistor connected at $t=135$ s, removed at $t=684$ s)

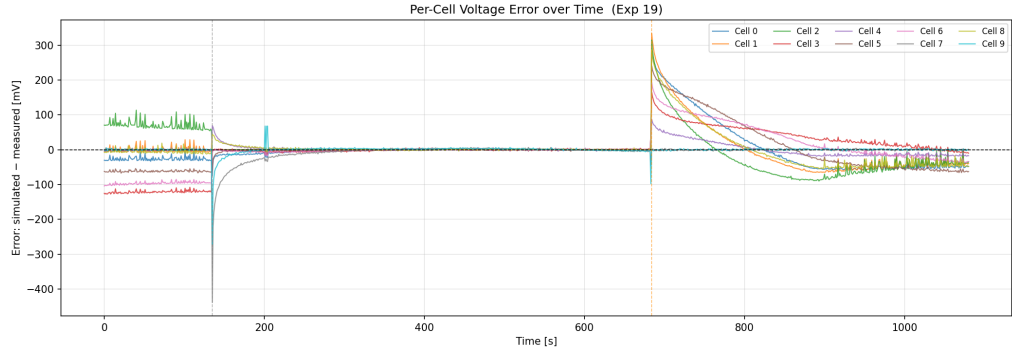


Load applied at $t = 135$ s via 25Ω resistor, removed at $t = 684$ s ($I = 0$ after removal).
Cells 0, 1, 2 & 8 collapse to ≈ 0 V (hardware degradation). Mean RMSE = 8.0 mV

Exp 19 — Stack Voltage, Current & Pressure



Exp 19 — Per-Cell Voltage Error



Large errors for degraded cells (0, 1, 2, 8) — model cannot capture hardware failure.

- Physical principles $\rightarrow$ semi-empirical voltage model $\rightarrow$ reduced state-space system
- 6 parameters tuned per cell using real experimental V-I data
- 8 out of 10 cells fitted with $\text{RMSE} \leq 5 \text{ mV}$
- Cell 2 (22 mV) and Cell 3 (10 mV) — degradation not fully captured by the model
- Concentration loss ($B > 0$) in Cells 1, 2, 7, 8, 9 $\rightarrow$ mass-transport limitations
- Parameters exported to `tuned_params.csv` for Simulink/MATLAB integration